Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_ASW")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```r
NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 =
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)


# text_priors <- '
# Distrib (pAm,       TruncNormal,      1172,    117.2, 800, 8000);
# Distrib (r_pAm_pM,  TruncNormal,      1.5,    0.2, 0.01, 100);
# Distrib (Fm,        TruncNormal,      0.3,    0.1, 0.1, 0.6);
# Distrib (kap,       Uniform,          0.05,        0.25);
# Distrib (Ehb, TruncNormal, 0.5, 0.5, 0.1, 10);
# Distrib (Ehp, TruncNormal, 100, 50, 1, 1000);
# # Distrib (alpha_Ehb, TruncNormal,      0.8,    0.1,    0, 1);
# # Distrib (beta_Ehp,  TruncNormal,        10,     1, 1, 1000);
# Distrib (kJ,        LogUniform,   0.00001,      0.1);
# Distrib (Eg,        TruncNormal,  6880.33,      1000, 0, 30000);
# Distrib (E_coc,     TruncNormal, 200, 100, 50, 5000);
#
# Distrib (muOM,      Normal,       11700,     500);
# Distrib (rOM_ClxHorse, TruncNormal,    0.2, 0.1, 0, 1);
# Distrib (v,         TruncNormal    0.02,     0.01, 0.000000000001, 100);
#
# # Calcul des vraissemblances
# Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_priors <- '
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40, 2000);
Distrib (Fm,           TruncNormal_cv,        0.3,      0.2,    0.01, 1);
Distrib (r_pAm_pM,     TruncNormal_cv,         0.8,      0.2,    0.01, 2);
Distrib (kap,          Uniform,                0.1,     0.9);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40, 300);
# Distrib (E_coc,        LogUniform,              10,      300);
Distrib (rOM_ClxHorse, TruncNormal,           0.01,      0.2,    0, 1);
Distrib (kJ,           TruncNormal_cv,       0.003,      0.2,    0.00001, 1);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'
```

```r
OM_diff <- TRUE

# makemcsim DEBcali.model

#f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbI

# ./mcsim.DEBcaliS DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? TRUE

3. Results

```
               pAm.1.      Fm.1. r_pAm_pM.1.      kap.1.    Ehp.1.
Result_mode 414.37700 0.28433800  0.12694200 0.899971000 126.02200
2.5%        391.66000 0.24981600  0.11040600 0.893793000 102.27530
97.5%       438.10475 0.32577500  0.15379960 0.899961000 160.24410
Min          89.15980 0.20235500  0.08758520 0.184565000  54.97850
Max         465.28400 0.36932500  0.79416300 0.899999000 200.67300
Result_mean 414.75750 0.28509318  0.13186967 0.898231766 130.17213
Result_sd    12.24909 0.01937505  0.01251085 0.004498973  14.90622
            rOM_ClxHorse.1.        kJ.1. Sigma_W.1.
Result_mode    2.819400e-03 0.0030012500  9.6323800
2.5%           2.958070e-04 0.0017574700  8.5925990
97.5%          2.427885e-02 0.0041340240 10.7498000
Min            1.922100e-07 0.0006224260  3.2454900
Max            2.826150e-01 0.0054420500 11.7522000
Result_mean    7.596584e-03 0.0029383344  9.6310069
Result_sd      6.663024e-03 0.0006030759  0.5468566

Number of observations, n =  363
Number of parameters, k =  7
Log-likelihood, LL =  -1168.83
BIC =  2378.921


Potential scale reduction factors:

                Point est. Upper C.I.
Ehp.1.                1.00       1.00
```

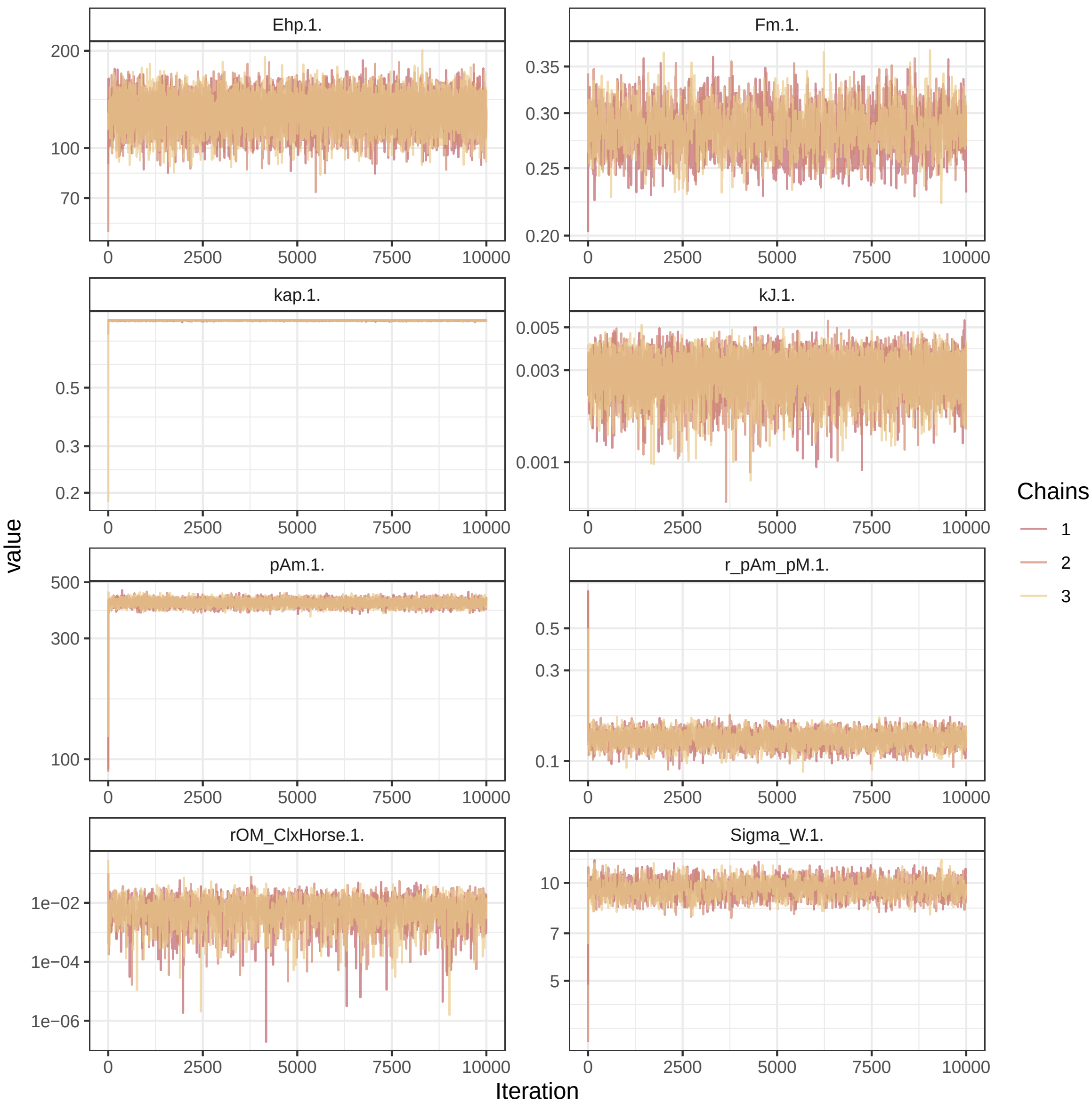


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

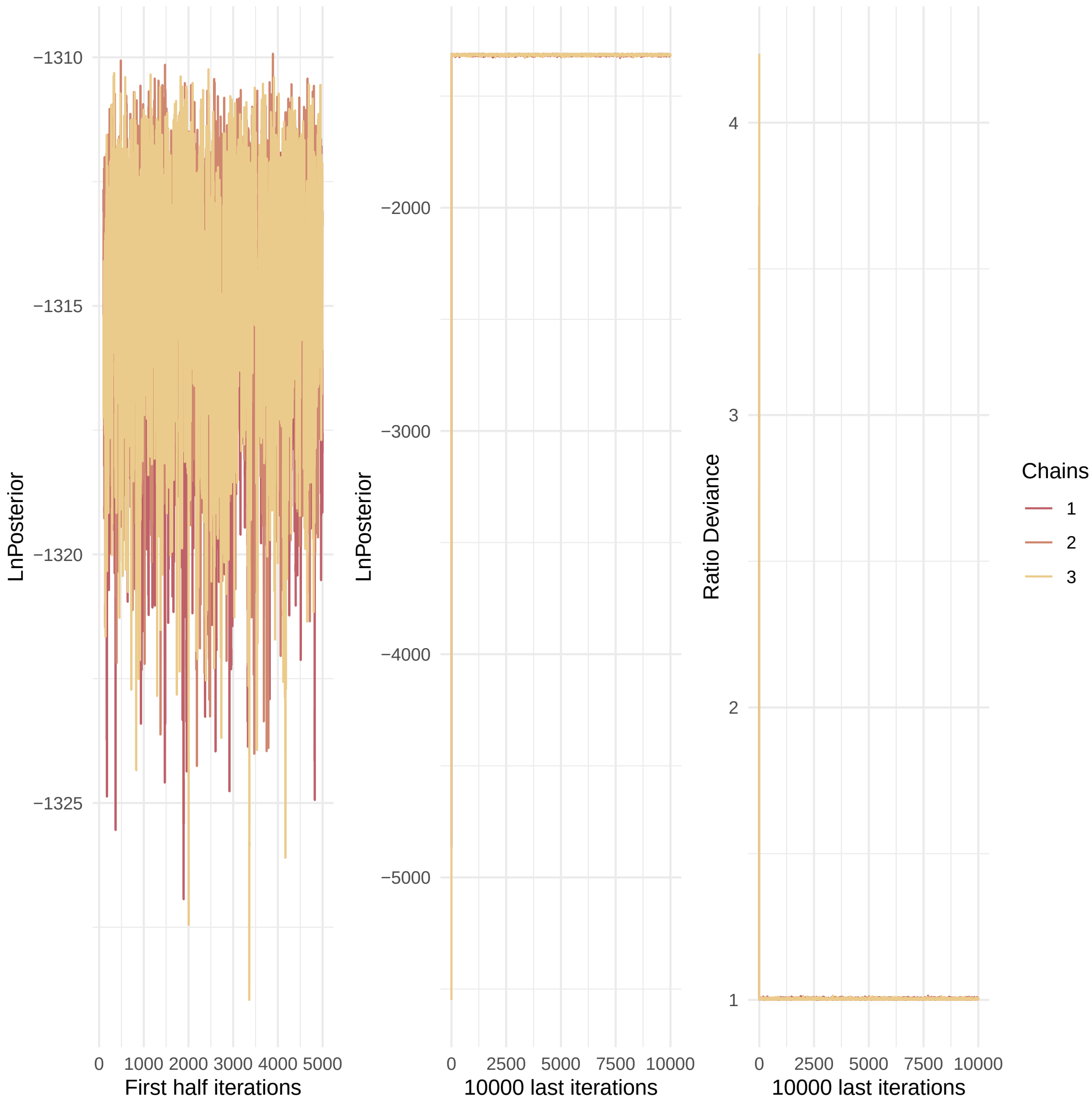


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

```
Fm.1.                  1.01       1.04
kap.1.                 1.00       1.00
kJ.1.                  1.00       1.01
pAm.1.                 1.00       1.01
r_pAm_pM.1.            1.00       1.01
rOM_ClxHorse.1.        1.00       1.01
Sigma_W.1.             1.00       1.01

Multivariate psrf

1.02
```

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

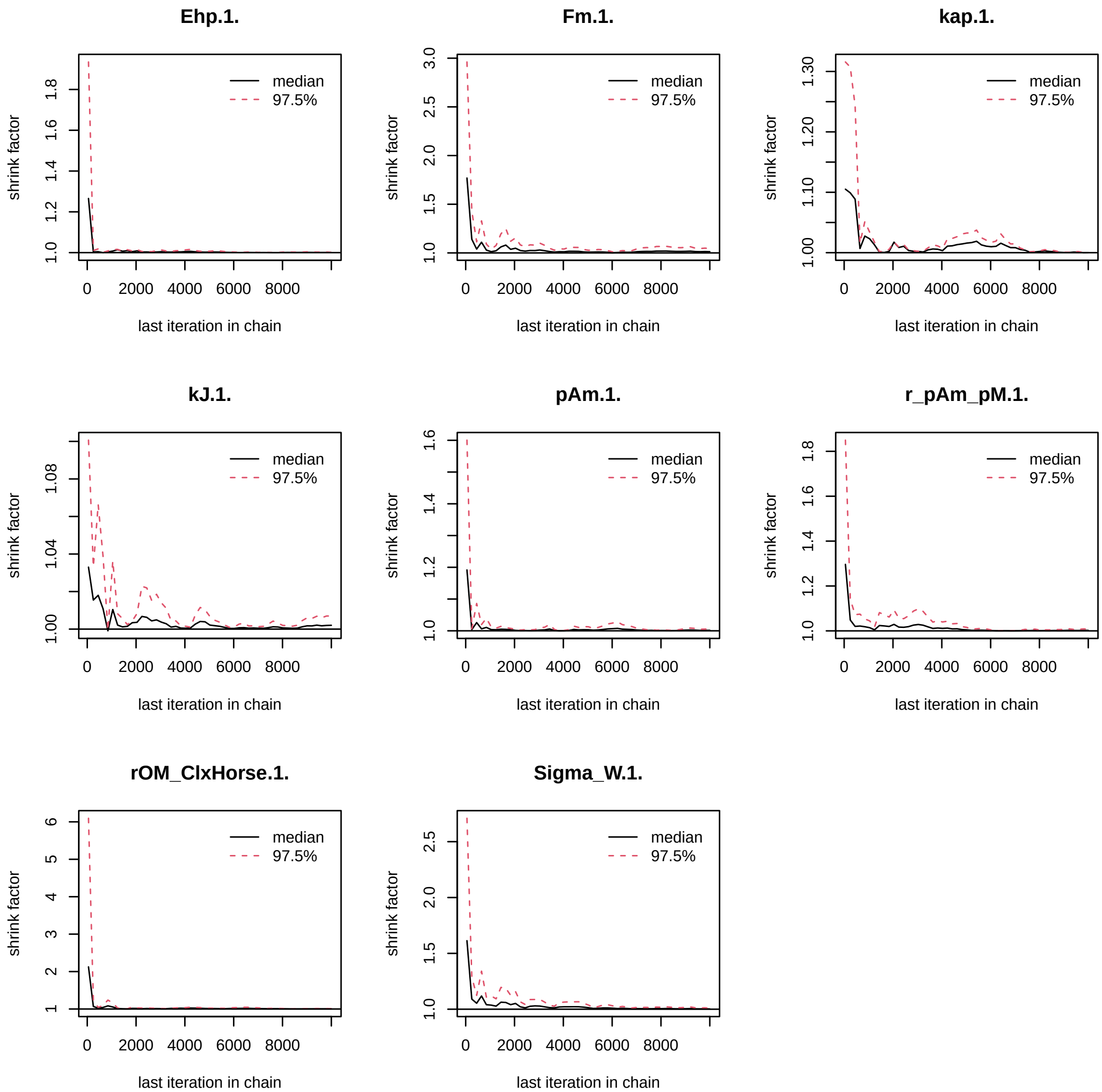


Figure 3: Chains convergence.

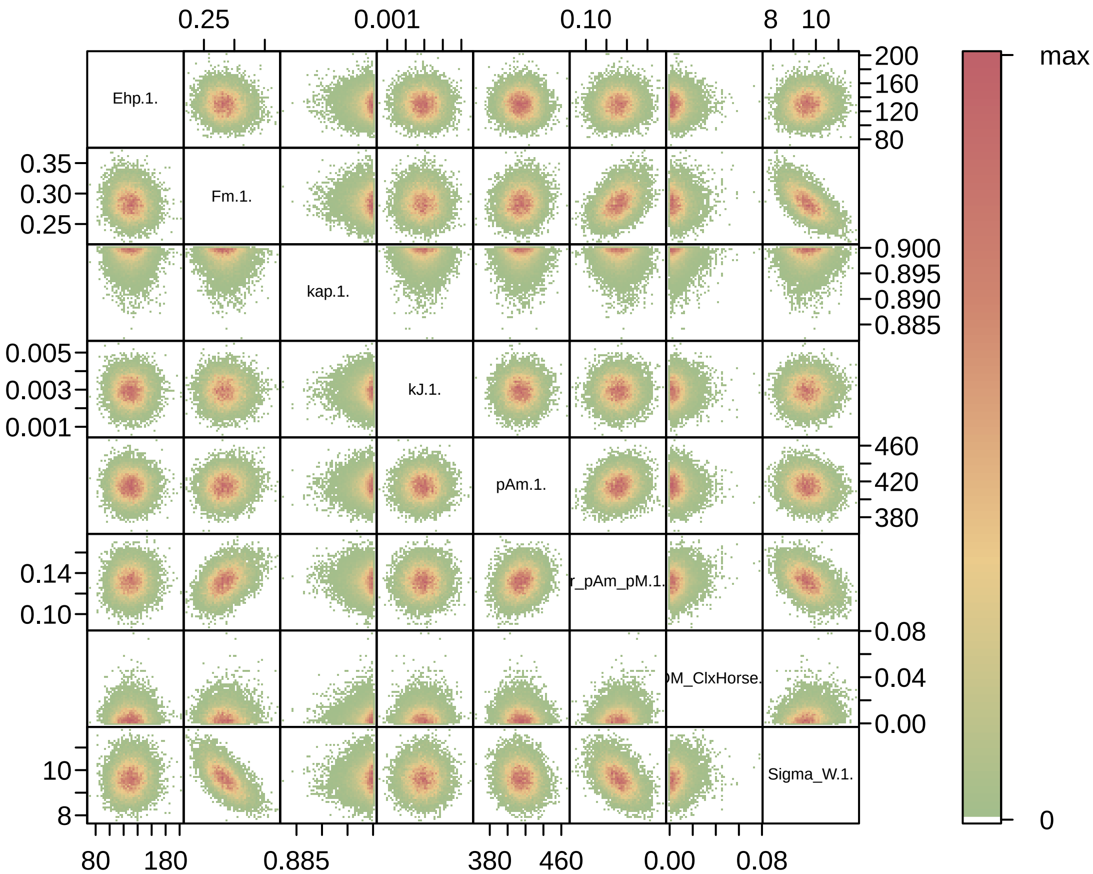


Figure 4: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

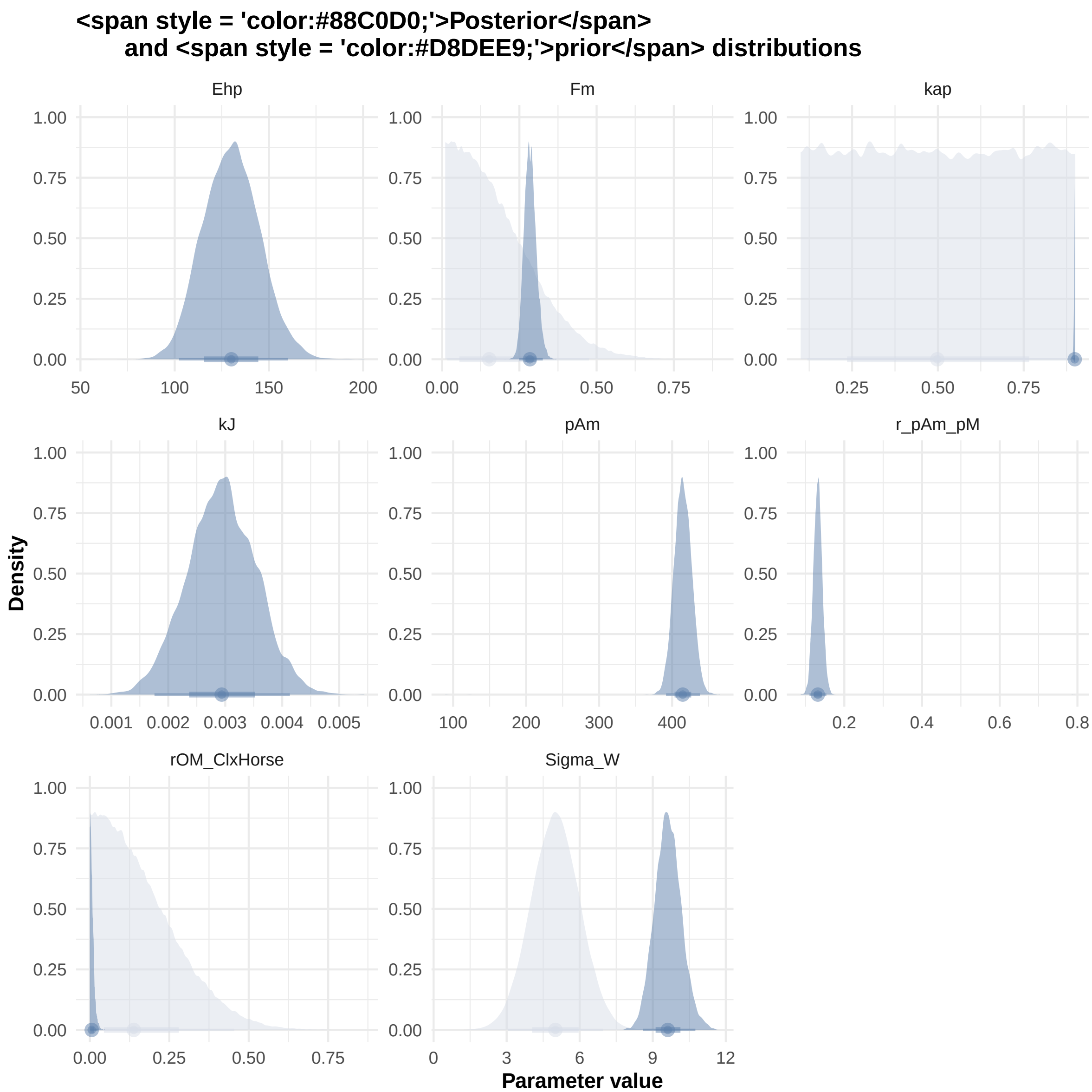


Figure 5: Distributions of the priors and the posteriors.

```
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)


text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Reproduction"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin

# ./mcsim.DEBcaliSW DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
```

```r
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

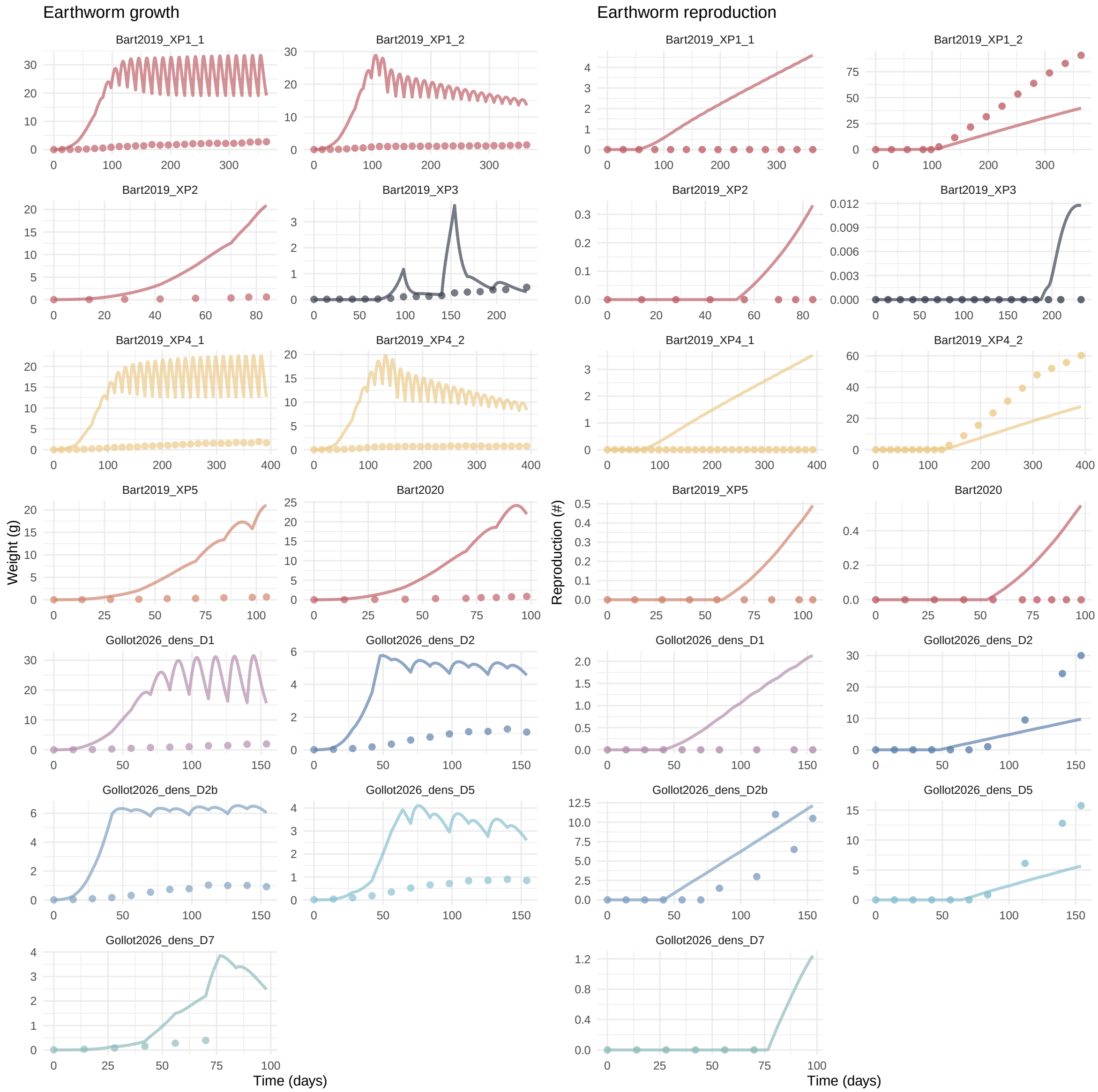
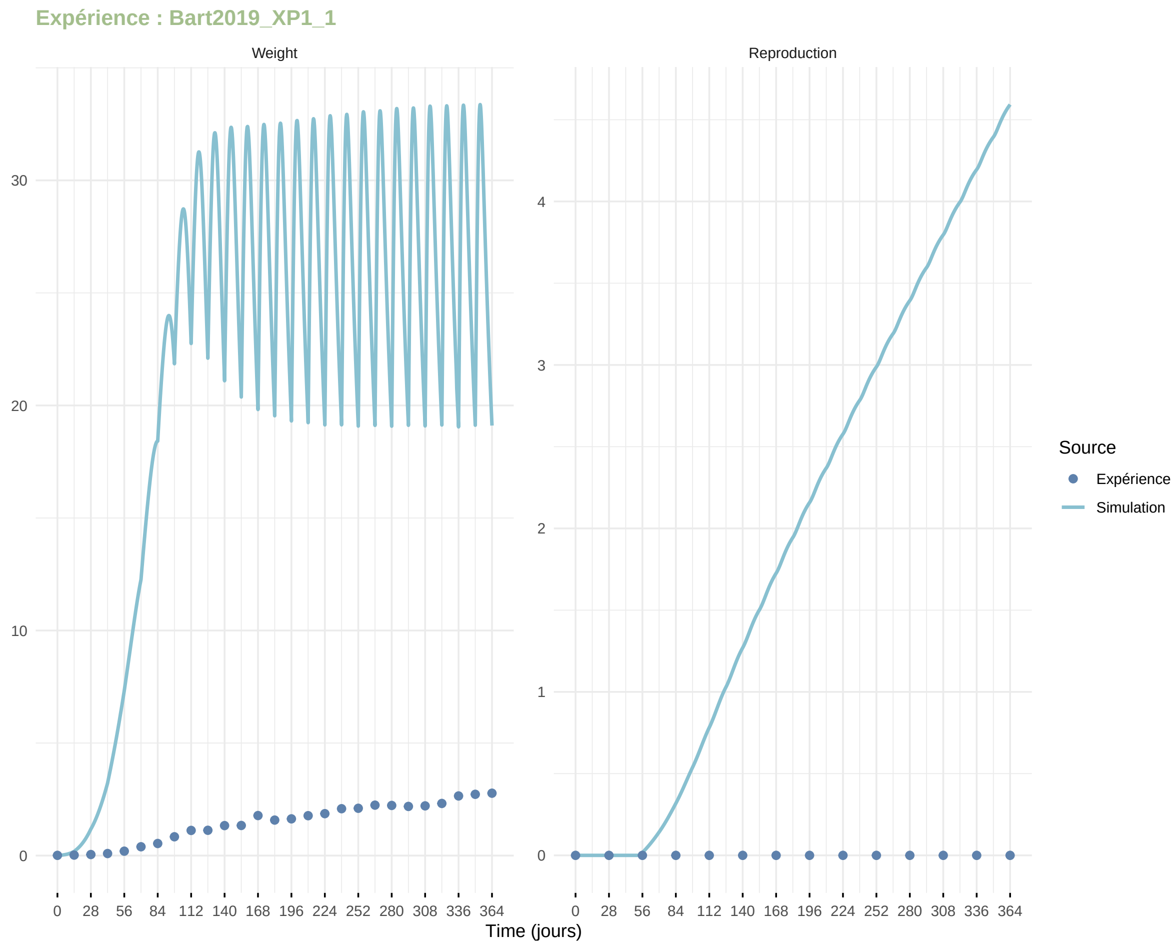
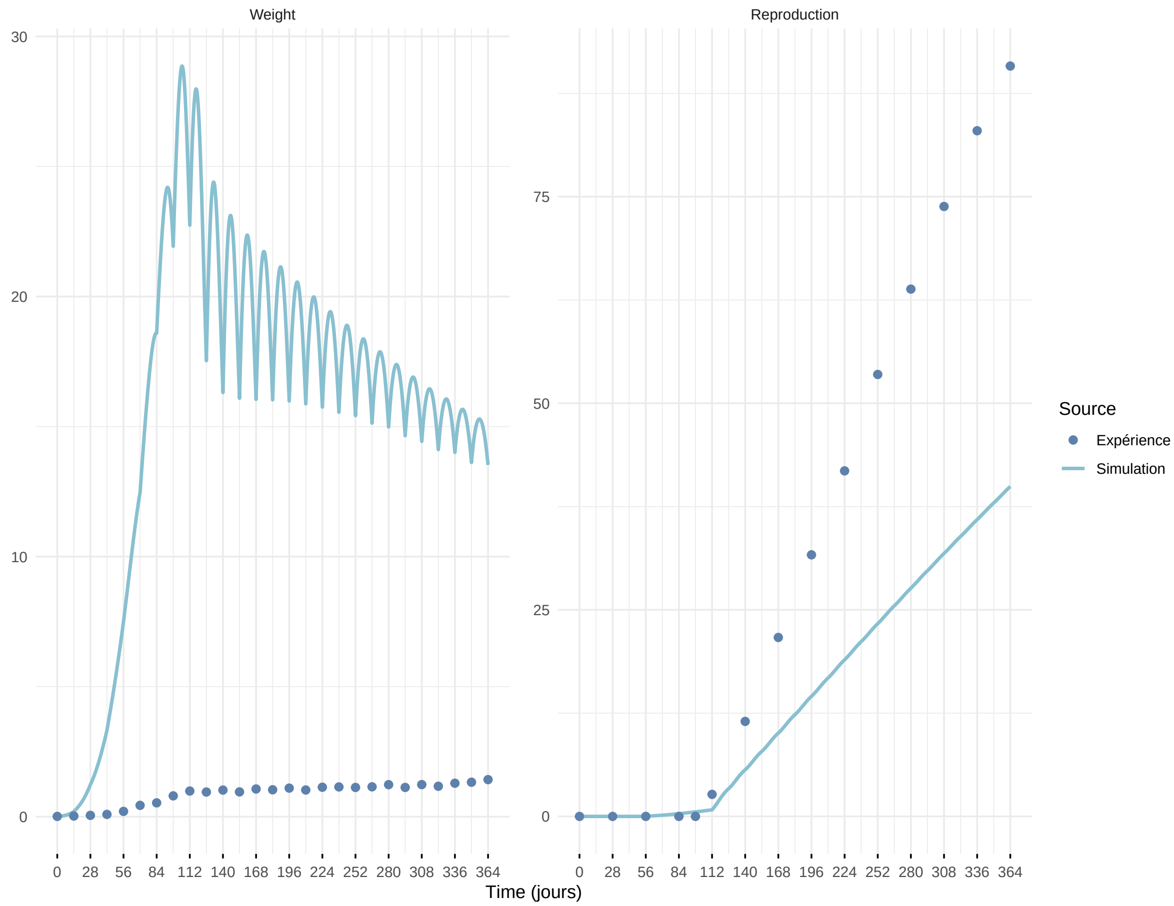


Figure 6: Simulations Weight and Reproduction.

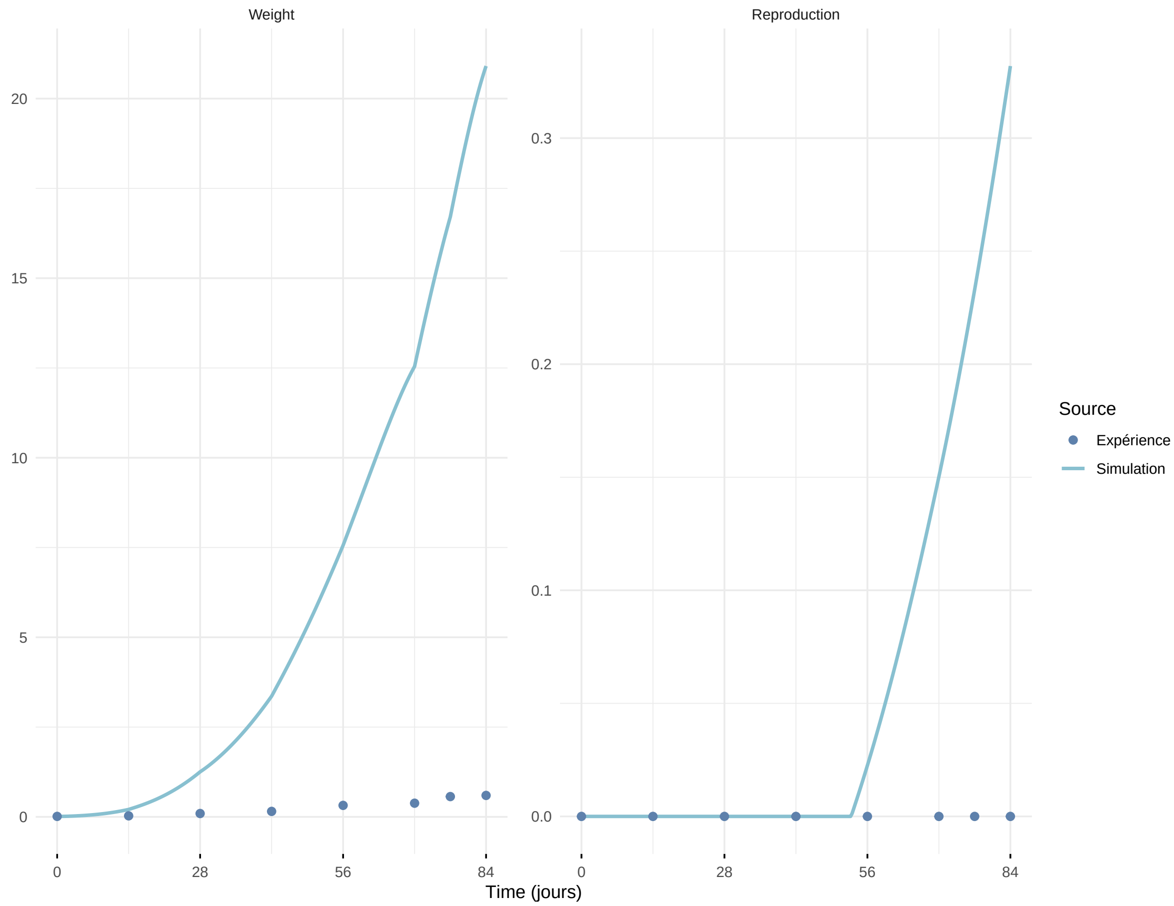


Expérience : Bart2019_XP1_1
Weight
Reproduction
Source
Expérience
Simulation
Time (jours)

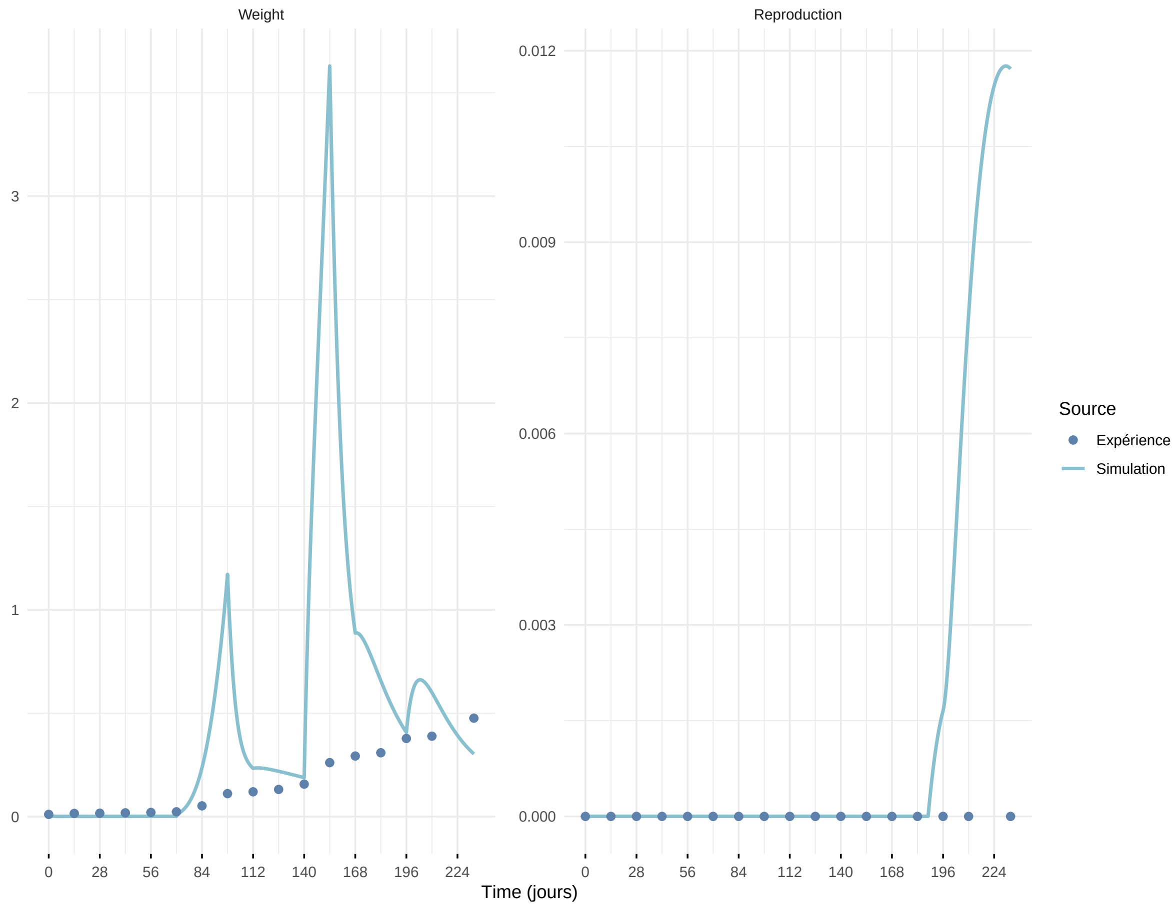
Expérience : Bart2019_XP1_2



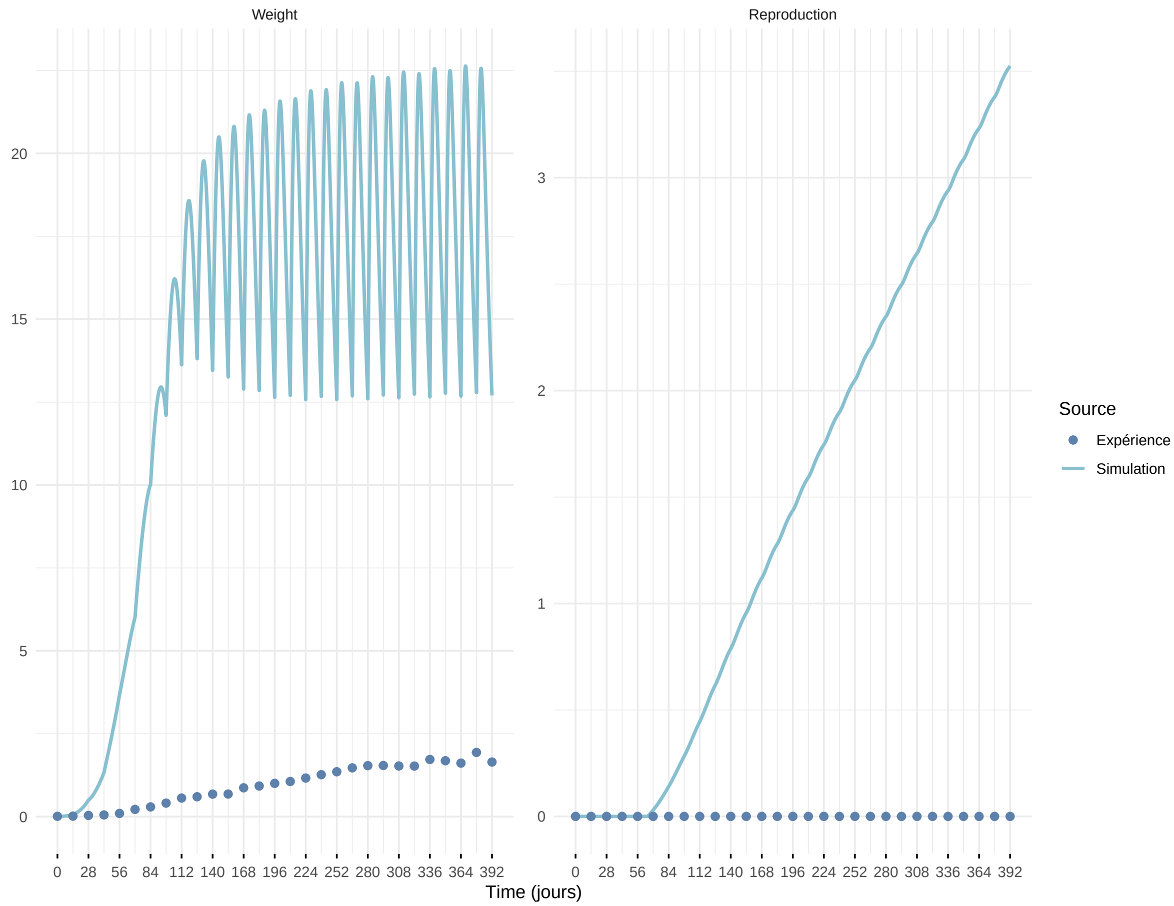
Expérience : Bart2019_XP2



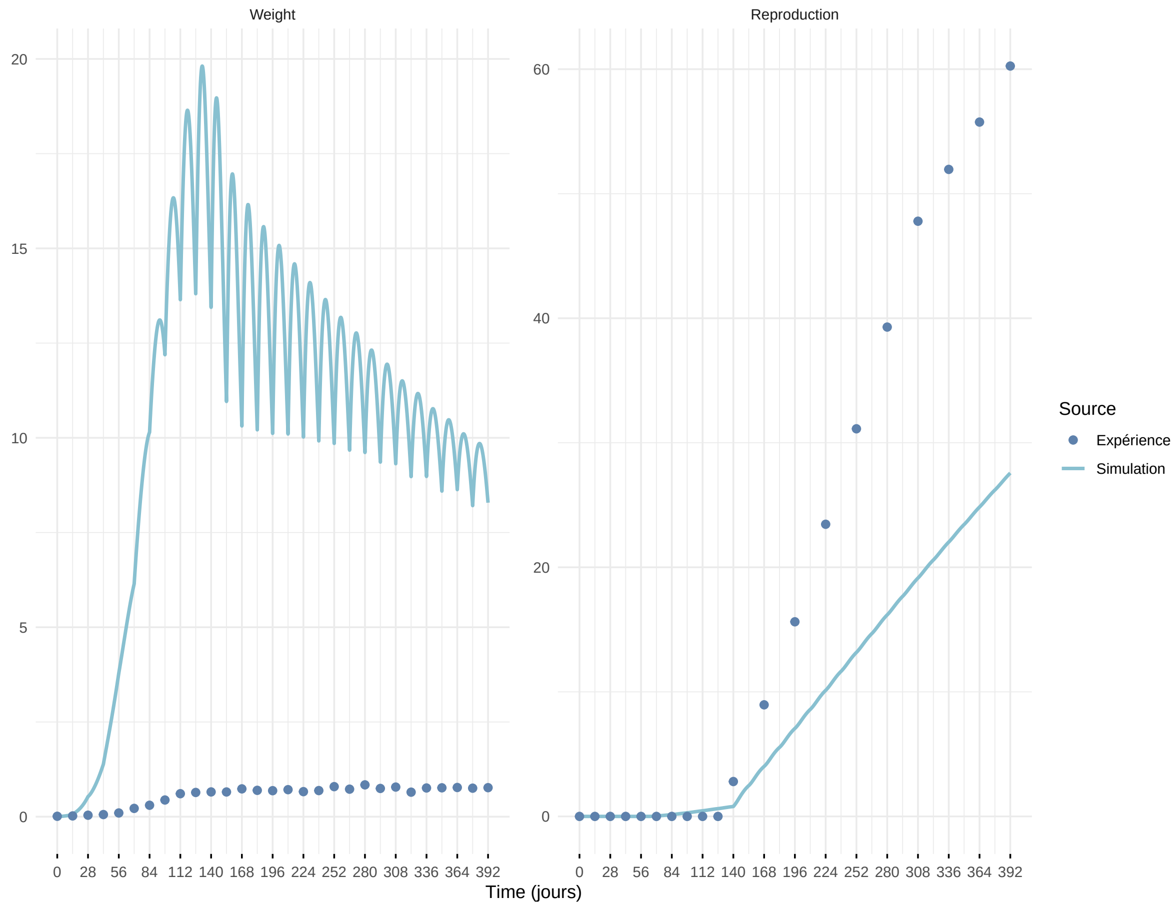
Expérience : Bart2019_XP3



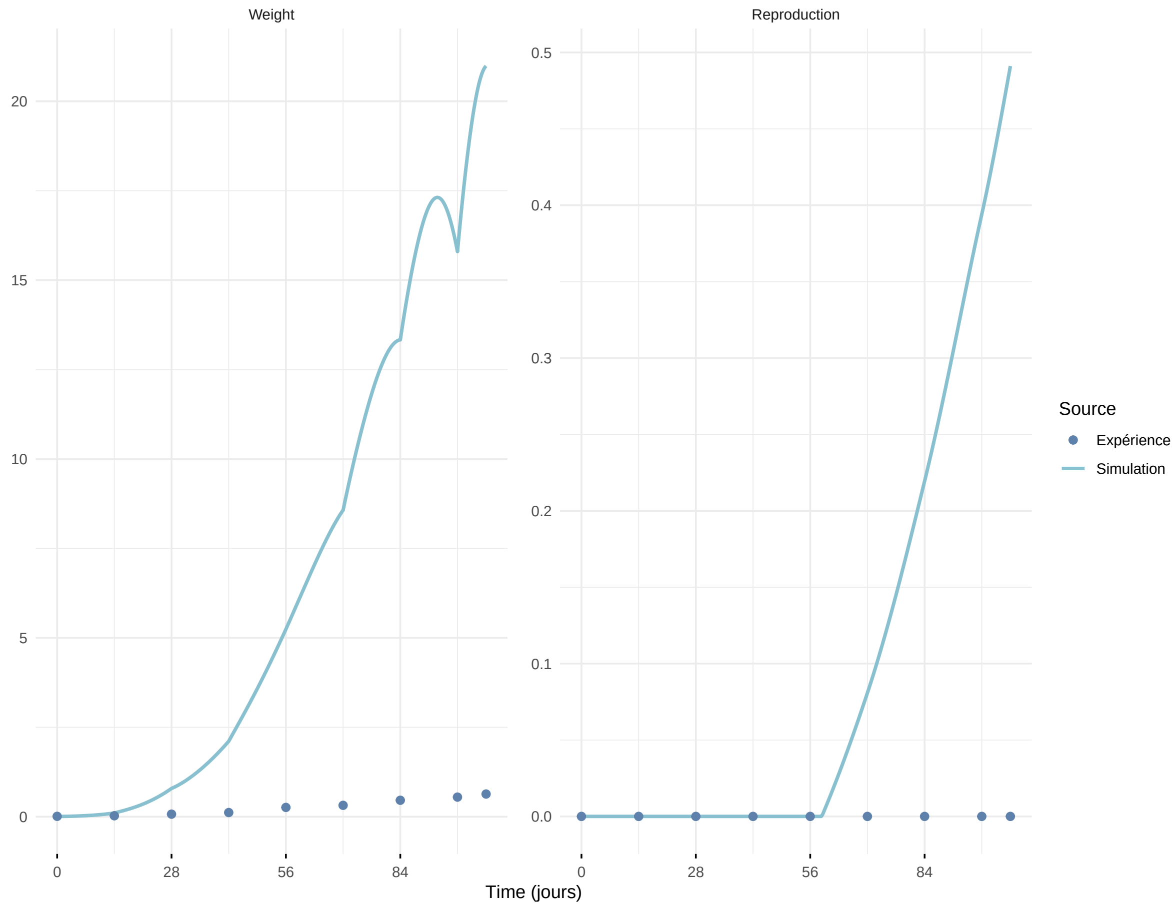
Expérience : Bart2019_XP4_1

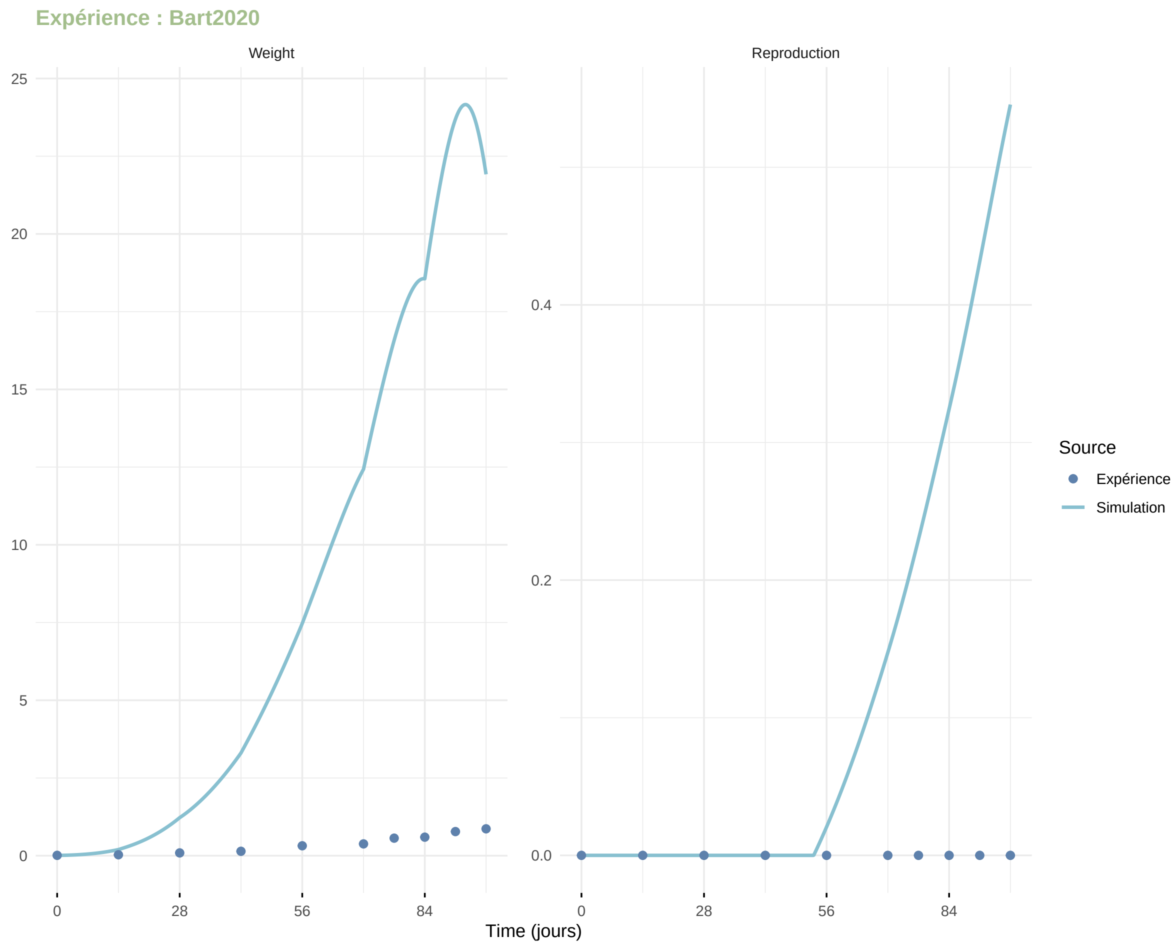


Expérience : Bart2019_XP4_2

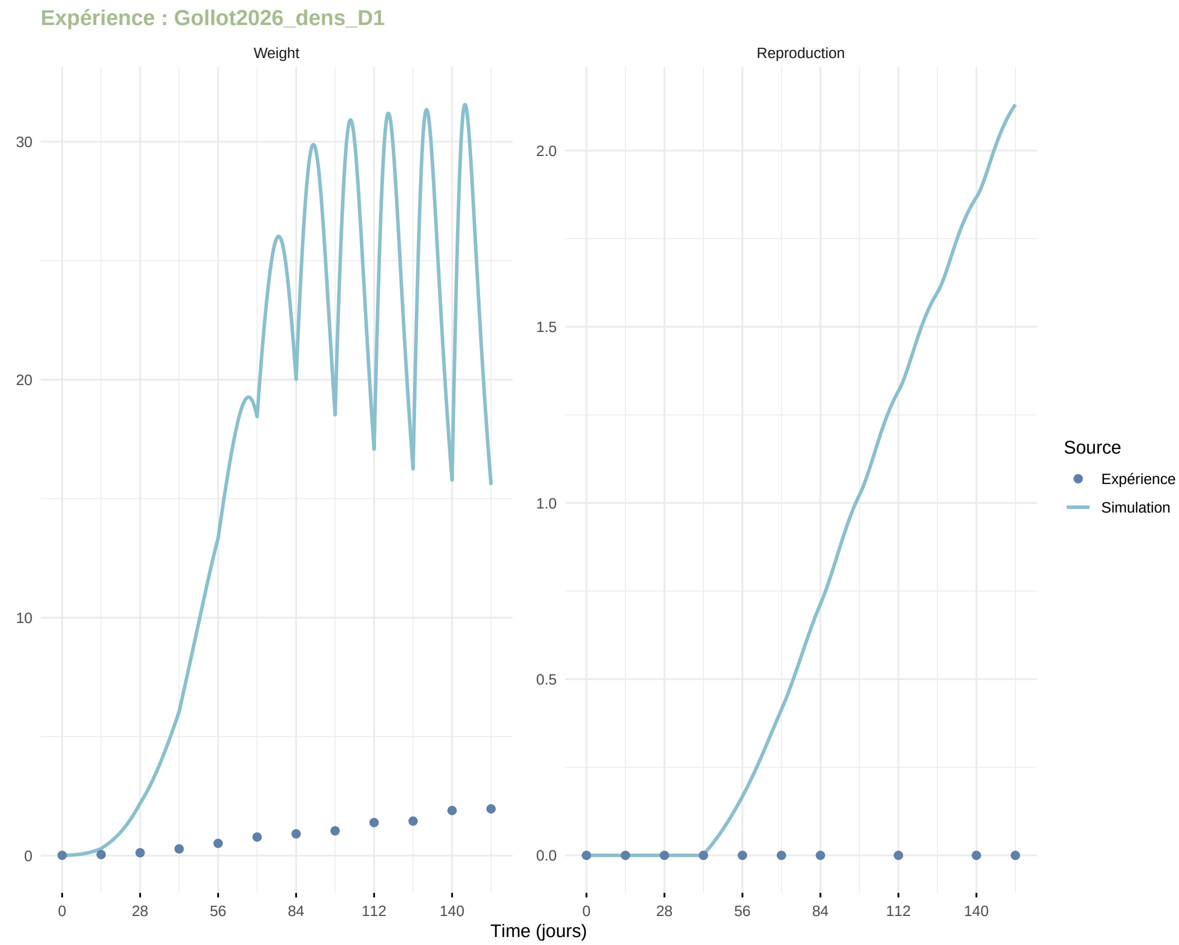


Expérience : Bart2019_XP5



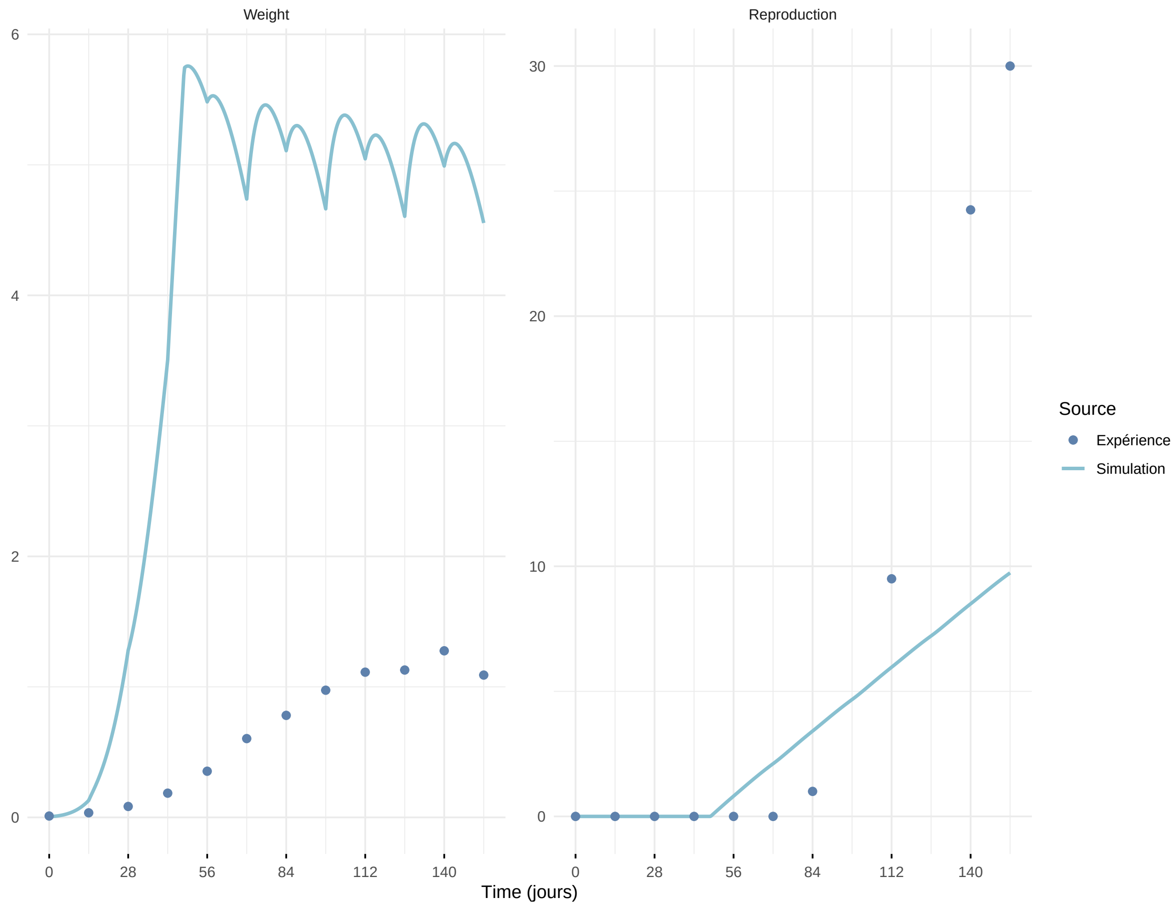


Expérience : Bart2020
Weight
Reproduction
25
20
15
10
5
0
0.4
0.2
0.0
0
28
56
84
Time (jours)
Source
Expérience
Simulation

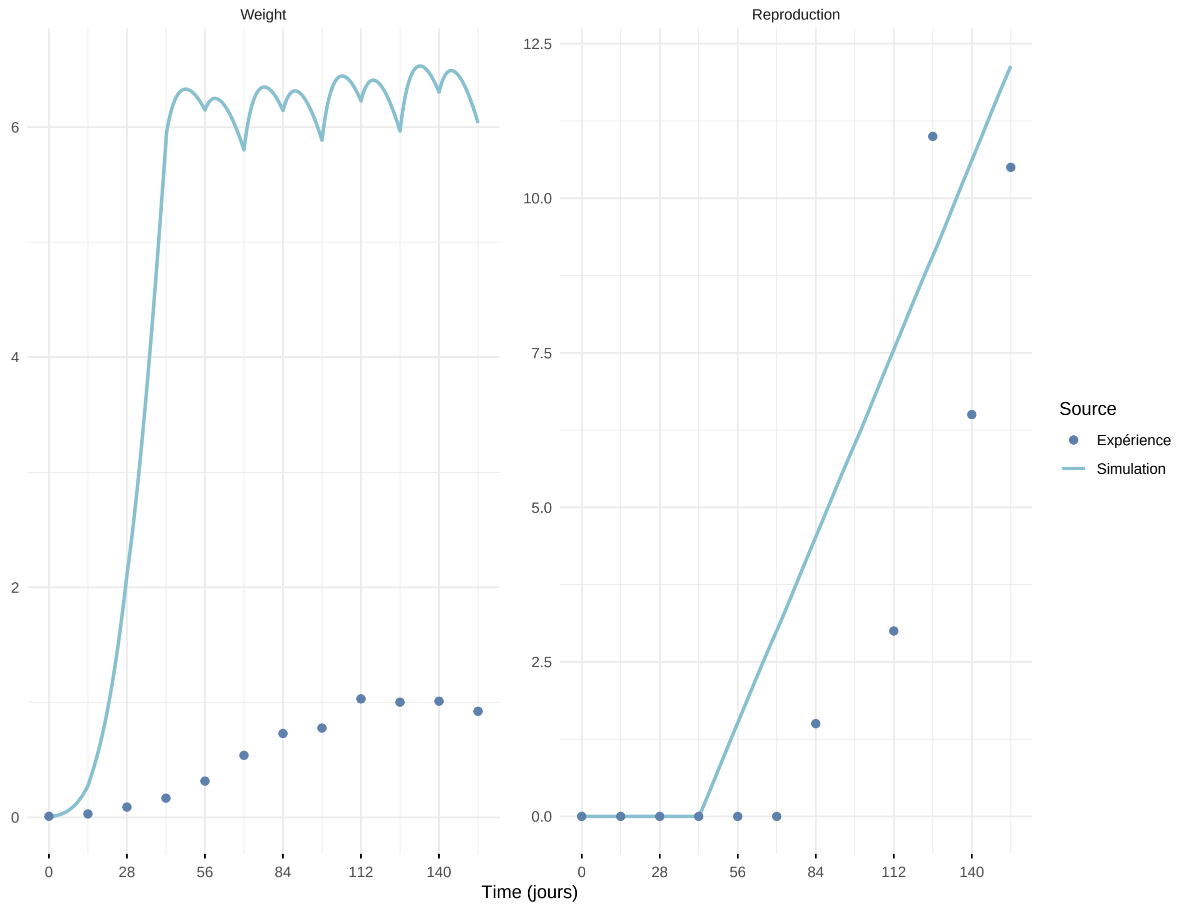


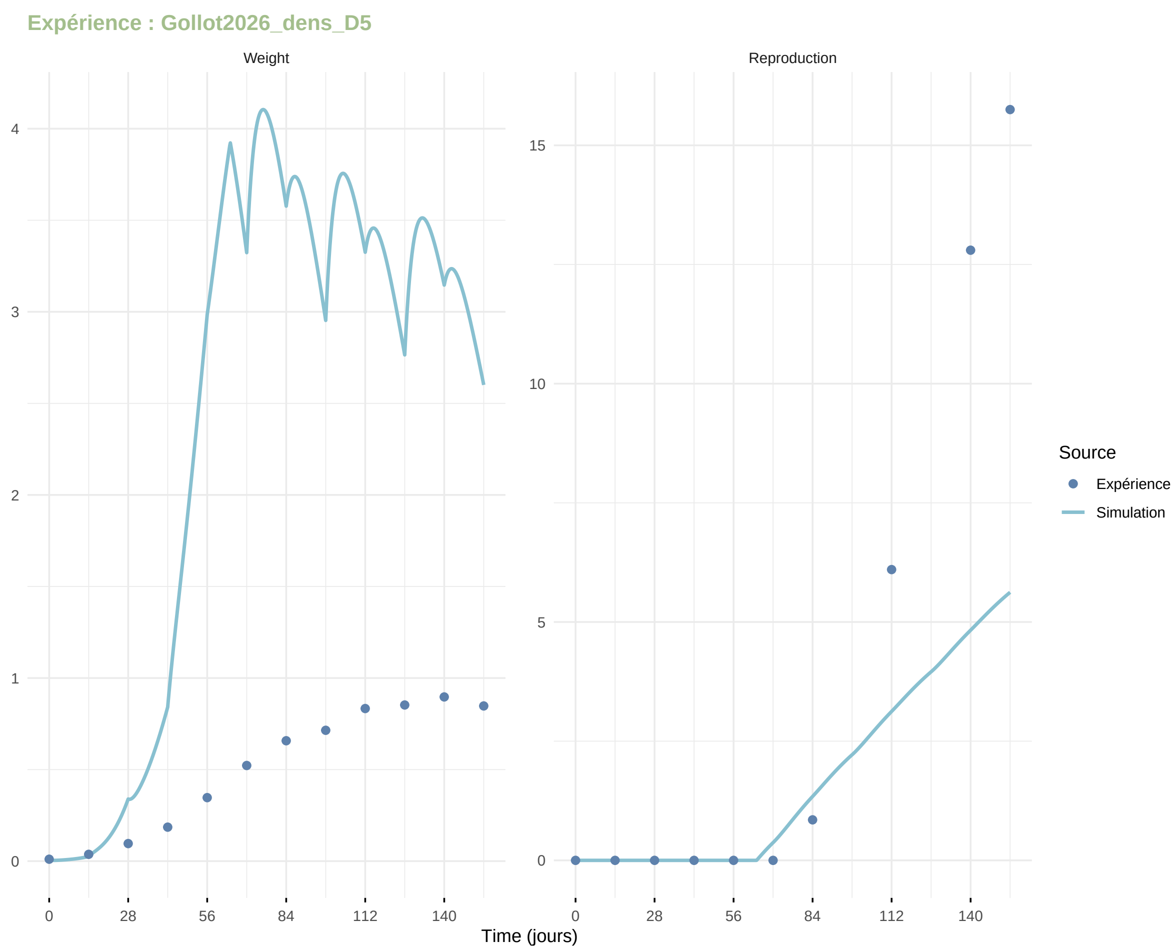
Expérience : Gollot2026_dens_D1
Weight
Reproduction
30
20
10
0
2.0
1.5
1.0
0.5
0.0
0
28
56
84
112
140
Time (jours)
Source
Expérience
Simulation

Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b





Expérience : Gollot2026_dens_D5
Weight
Reproduction
Source
Expérience
Simulation
Time (jours)

Expérience : Gollot2026_dens_D7